

Every week, Paulie Walnuts makes his rounds collecting tribute envelopes for the family. Tony wants a read on Paulie's average weekly take (λ). Is he skimming, or is business just slow?

Consider the Gamma-Poisson model for λ with

$$Y_i | \lambda \sim \text{Pois}(\lambda)$$

and

$$\lambda \sim \text{Gamma}(20, 5)$$

Suppose Tony has Silvio keep tabs on Paulie for $n=3$ weeks, observing $(Y_1, Y_2, Y_3)=(0, 1, 0)$

a. Simulate the posterior model of λ with RStan using 4 chains and 10,000 iterations per chain.

Python code:

```
# pip install pystan

import stan # note: PyStan 3.x import is `stan`, not `pystan`

import matplotlib.pyplot as plt

import numpy as np

from distributions import gamma_pdf

stan_code = """

data {
  int<lower=0> N;

  array[N] int<lower=0> Y;
}

parameters {
  real<lower=0> pi;
}

model {
  Y ~ poisson(pi);
}
```

```

pi ~ gamma(20, 5);

}

"""

actual_iterations = 5_000

chains = 4

data = {"N":3,

        "Y": (0,1,0)}

posterior = stan.build(stan_code, data=data, random_seed=84735)

fit = posterior.sample(num_chains=chains, num_samples=actual_iterations,
num_warmup=actual_iterations)

pi_draws = fit["pi"].reshape((chains,actual_iterations)) # numpy array, shape (1,
num_samples * num_chains) since pi is scalar

pi_draws_flat = pi_draws.reshape(1,-1)

chains = {

}

colors = ["#B0C4DE", "#A8B8C8", "#C0CCD9", "#7B92A8"]

# Caterpillar

for i,c,color in zip(np.arange(1,5),pi_draws,colors):

    chains.setdefault(str(i),c)

    plt.plot(np.arange(1,actual_iterations+1),c,label=i,color=color)

plt.legend()

```

```

plt.show()

# Histogram
for i,c,color in zip(np.arange(1,5),pi_draws,colors):

    plt.hist(pi_draws_flat[0],70,density=True,color=color,label=i)

plt.legend()

plt.xticks(np.arange(0,7,0.5))

plt.show()


xs = np.linspace(0,7,5000)

ys = np.array([gamma_pdf(x,20,1/5) for x in xs])

plt.plot(xs,ys)

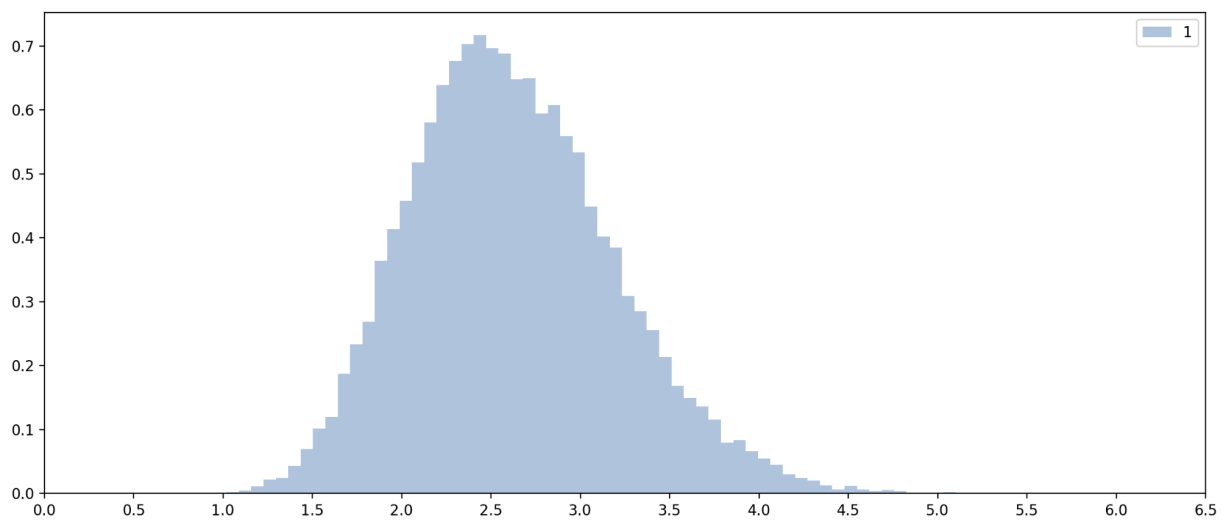
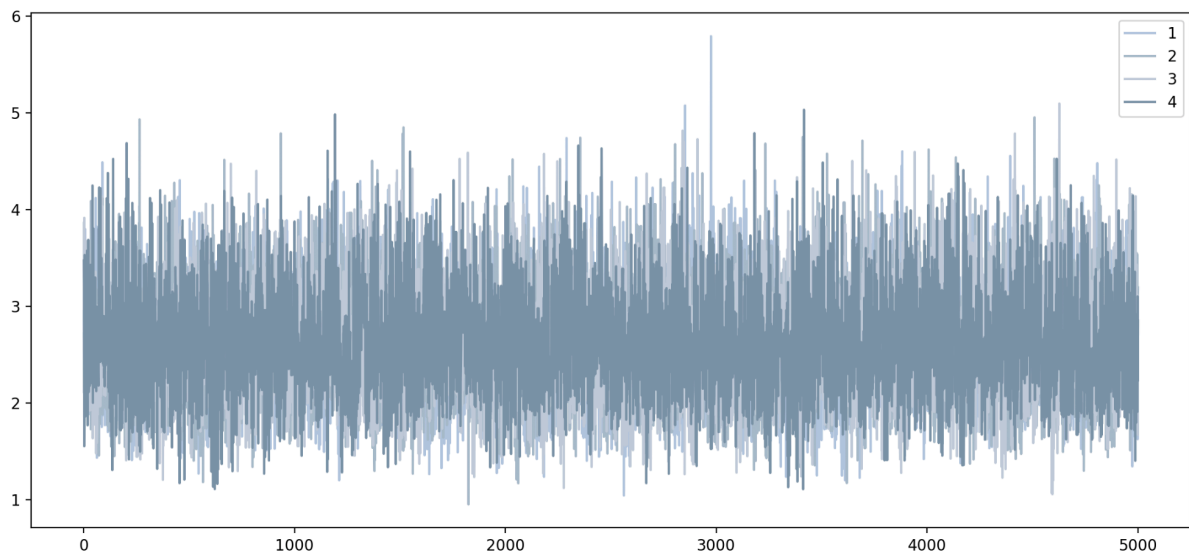
plt.legend()

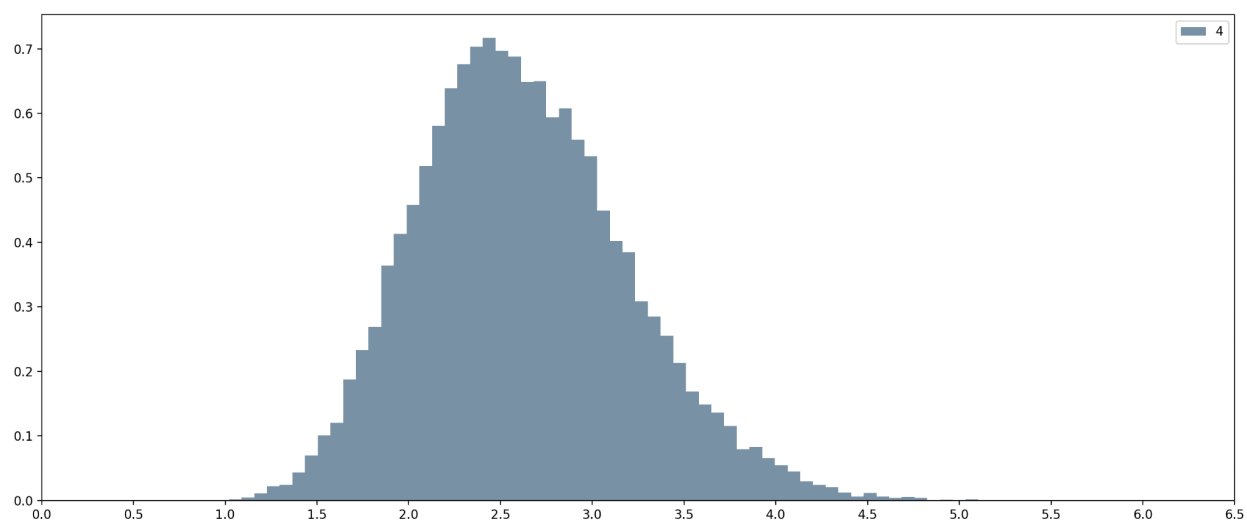
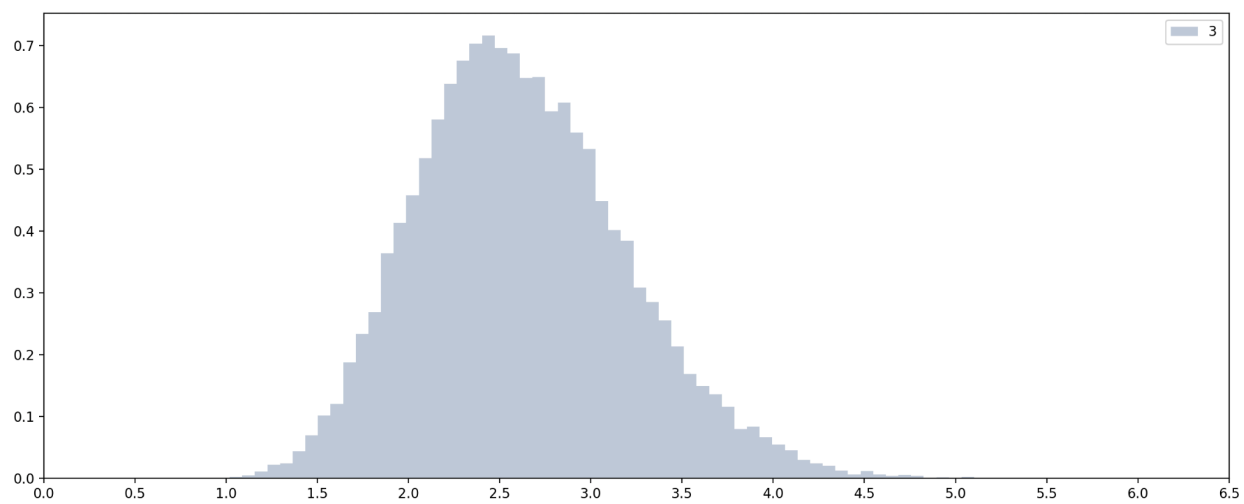
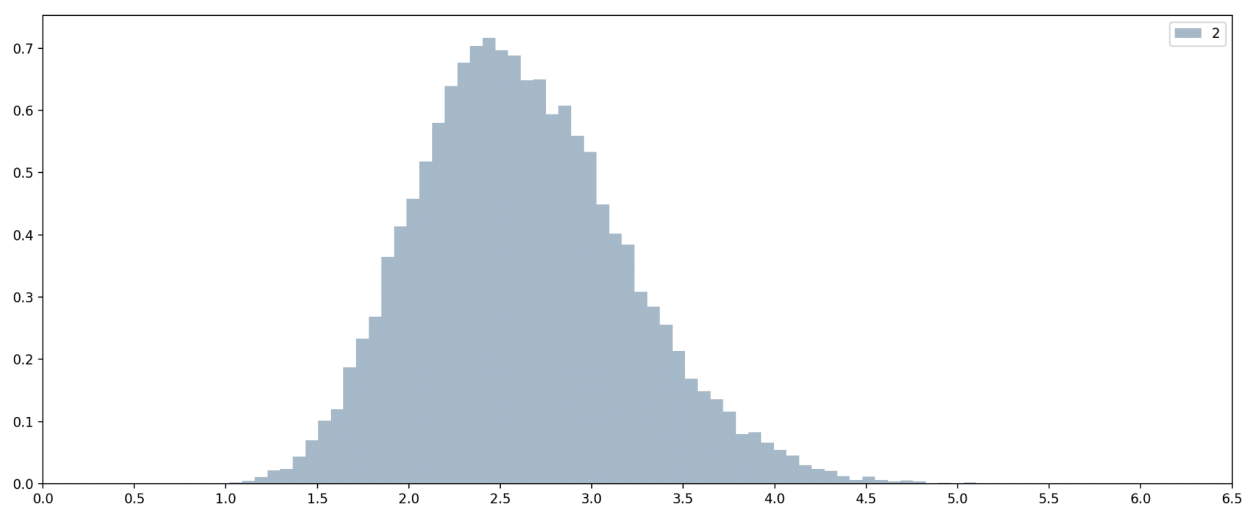
plt.title("Known Posterior Model of lambda")

plt.show()

```

b. Produce trace and density plots for all four chains.

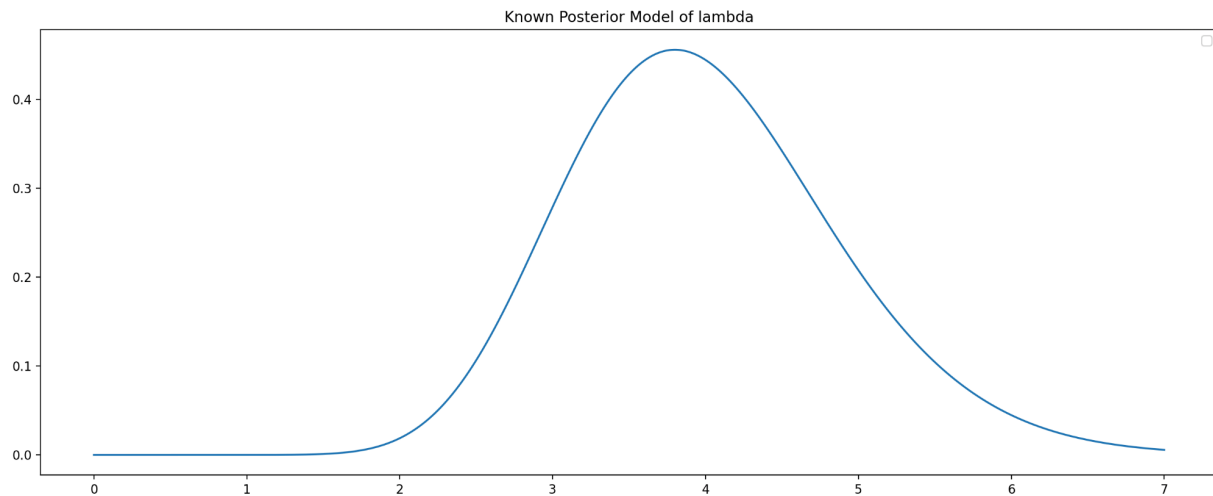




c. From the density plots, what seems to be the most posterior plausible value of λ ? (Should Tony be worried?)

From the density plots, it looks like the most plausible value of λ is about 2.5

d. Going back to Chapter 5, specify the known posterior model of λ analytically. How does your MCMC approximation compare?



The known posterior model of λ is a Gamma(20,5) model, which puts the most likely value of λ between 3.5 and 4.